

AVI BENJAMIN SHEIN

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EDUCATION

Northwestern University, TGS, Evanston, IL

September 2024 - June 2026

Candidate for Master of Science in Computer Science; GPA: 3.97

Relevant Courses: Graduate Algorithms, Approximation Algorithms, Graduate Cryptography, Theory of Gradient Based Optimization, Algorithms for Collective Decision Making, Quantum Computing

Tufts University, School of Engineering, Medford, MA

May 2024

B.S. in Computer Science, Math Minor; GPA: 3.9 (*summa cum laude*), 6 semester TA (Algorithms, Discrete)

Relevant Courses: Computational Geometry, Graph Theory, Theory of Computation, Abstract Algebra

EXPERIENCE

EndoSec LLC (Security and Cryptography), *Software Engineering Intern*

June - September 2025

- Created a CI/CD automated workflow for FPGA engineers to help streamline synthesis, simulation, and automated testing
- Developed a virtual environment for engineers to standardize software and hardware production

Research on the BPE Algorithm, (Prof. Konstantin Makarychev), *Researcher*

January 2025 - Present

- Analyzed algorithms for the tokenization step of LLMs such as ChatGPT
- Constructed proofs to improve upper and lower bounds of the Byte Pair Encoding (BPE) algorithm

NSF REU Research in Computational Geometry, (Prof. Csaba Toth), *Researcher*

June - August 2023

- Developed spanner proofs which are used in computational geometry to solve proximity problems
- Focused on creating spanners of intersection graphs with fewest edges in a given hop distance
- Created upper and lower bound proofs for 2-hop spanners of wedges, unit squares, and unit cubes

8vdX, *Software Engineering Intern*

June - August 2022

- Built dynamic charts and forms for displaying Key Performance Indicators for 8vdX, an India-based, Y Combinator funded fintech company, that matched start-ups with venture debt investors

Technion Department of Computer Science, (Prof. Gershon Elber) *Researcher*

June - August 2021

- Created customizable semiregular geometric tiles used to make 3D printed objects
- Published as 2nd author on paper titled "[Shell-lattice construction based on regular and semi-regular tiling via functional composition](#)"

PROJECTS

Project Hybrot

Fall 2023 - Spring 2024

- Developed a program and GUI for a computational biology group (Levin Lab) at Tufts University
- Created software to stimulate neurons in a closed loop to observe how they learn information, set up and record experiments, and analyze resulting data

Terminal Chess

Winter 2022 - Summer 2023

- Programmed a C++ [chess game](#) played in Terminal with customizable board and pieces
- Implemented game variants, and created chess bots to play against as opponents

SKILLS & INTERESTS

Languages: C, C++, C#, JavaScript, Python, Java, HTML, CSS, x86 ASM

Software Programs/Tools: Shell Scripting, Unity, GitHub, Bitbucket, ReactJS

Interests: Puzzles, Chess, Rubik's Cubes, Premier League (Chelsea), Fantasy (TV series/Books), Baking